

PART 2 — LEVEL 2 (BASIC MCQs)

SECTION 1: Electric Charges & Properties (Q1–10)

- Which of the following is a property of electric charge?
 - It is continuous
 - It is conserved
 - It can be created
 - It cannot be transferred
- The SI unit of electric charge is:
 - Volt
 - Coulomb
 - Ampere
 - Farad
- Two bodies attract each other. The bodies are:
 - Both positive
 - Both negative
 - One positive, one negative
 - Neutral
- Which of the following is always true for an isolated system?
 - Total charge remains zero
 - Total charge remains constant
 - Charges get neutralized
 - Charges cannot move
- What happens when a rubber rod is rubbed with fur?
 - Rubber gains electrons
 - Rubber loses electrons
 - Fur gains protons
 - No charge is generated
- Which instrument is used to detect electric charge?
 - Voltmeter
 - Ammeter
 - Electroscope
 - Galvanometer
- Which of the following is a conductor?
 - Rubber
 - Copper
 - Plastic
 - Glass
- Which charge is carried by an electron?
 - Positive
 - Negative
 - Neutral
 - None of the above
- The principle of conservation of charge was first stated by:
 - Coulomb

- B) Benjamin Franklin
 - C) Faraday
 - D) Maxwell
10. Which of the following statements is correct?
- A) Charge can appear spontaneously
 - B) Charge can disappear spontaneously
 - C) Net charge in an isolated system is conserved
 - D) Net charge always becomes zero

◆ SECTION 2: Conductors, Insulators & Induction (Q11–20)

11. Which of the following is an insulator?
- A) Copper
 - B) Aluminium
 - C) Glass
 - D) Silver
12. Charging by friction occurs due to:
- A) Electron transfer
 - B) Proton transfer
 - C) Neutron transfer
 - D) Positron transfer
13. In charging by induction, the object:
- A) Touches the charged body
 - B) Does not touch the charged body
 - C) Always becomes neutral
 - D) Gains mass
14. The electric field inside a conductor in electrostatic equilibrium is:
- A) Maximum
 - B) Zero
 - C) Negative
 - D) Infinite
15. A hollow metallic sphere is charged. The charge resides:
- A) On inner surface
 - B) On outer surface
 - C) Uniformly inside
 - D) Distributed randomly
16. Earthing is done to:
- A) Increase charge
 - B) Prevent charge accumulation
 - C) Change polarity
 - D) Generate current
17. Which of the following statements is true?
- A) Conductors have fixed electrons
 - B) Insulators have free electrons
 - C) Conductors allow electron movement
 - D) Conductors do not allow electrons

18. A charged comb attracts small paper pieces due to:
A) Induction
B) Polarization
C) Conduction
D) Friction
19. Why does a sharp point on a conductor leak charge easily?
A) Low potential
B) High potential gradient
C) Low field
D) Conductor property
20. A material which partially conducts electricity is:
A) Conductor
B) Insulator
C) Semiconductor
D) Dielectric

◆ SECTION 3: Coulomb's Law (Q21–30)

21. Coulomb's law states that force between two charges is:
A) Directly proportional to distance
B) Inversely proportional to square of distance
C) Independent of distance
D) Directly proportional to square of distance
22. The electrostatic force is:
A) Always attractive
B) Always repulsive
C) Attractive or repulsive
D) Zero
23. If the distance between two charges is doubled, the force becomes:
A) 4 times
B) 2 times
C) $1/4$
D) $1/2$
24. The SI unit of Coulomb's constant is:
A) $\text{N}\cdot\text{m}^2/\text{C}^2$
B) $\text{N}\cdot\text{m}/\text{C}$
C) $\text{V}\cdot\text{m}$
D) J/C^2
25. Coulomb's law is valid:
A) For all charges
B) Only for point charges
C) For magnetic poles
D) For moving charges only
26. Two equal positive charges are placed 1 m apart. Force is F . If one charge is doubled, force becomes:
A) F
B) $2F$

- C) $F/2$
 - D) $4F$
27. Electrostatic force is a:
- A) Vector
 - B) Scalar
 - C) Tensor
 - D) Matrix
28. The principle of superposition is used to:
- A) Calculate net force
 - B) Calculate work
 - C) Calculate potential
 - D) Measure current
29. Which of these affects Coulomb force?
- A) Magnitude of charges
 - B) Distance between charges
 - C) Medium between charges
 - D) All of the above
30. Two point charges attract each other with 36 N. If each charge is halved, the new force is:
- A) 9 N
 - B) 18 N
 - C) 36 N
 - D) 72 N

◆ SECTION 4: Electric Field & Field Lines (Q31–50)

31. Electric field at a point is:
- A) Force per unit charge
 - B) Work per unit charge
 - C) Energy per unit charge
 - D) Charge per unit distance
32. SI unit of electric field is:
- A) N/C
 - B) C/N
 - C) V/m^2
 - D) $V \cdot m$
33. Electric field lines start from:
- A) Negative charges
 - B) Positive charges
 - C) Neutral bodies
 - D) Dipoles
34. Electric field lines end at:
- A) Positive charges
 - B) Negative charges
 - C) Neutral points
 - D) Infinity
35. Which is true about field lines?
- A) They intersect

- B) They never intersect
 - C) They are discontinuous
 - D) They form loops
36. Field inside a uniformly charged spherical shell is:
- A) Maximum at center
 - B) Zero
 - C) Constant non-zero
 - D) Varies linearly
37. Electric field due to point charge decreases as:
- A) $1/r$
 - B) $1/r^2$
 - C) $1/r^3$
 - D) Independent of r
38. A test charge is:
- A) Very large
 - B) Very small
 - C) Negative
 - D) Positive
39. Field lines for dipole:
- A) Radial only
 - B) Straight
 - C) Curved
 - D) Parallel
40. Uniform electric field is produced between:
- A) Parallel plates
 - B) Point charges
 - C) Ring charge
 - D) Spherical shell
41. Tangent to electric field lines gives:
- A) Direction of field
 - B) Magnitude
 - C) Potential
 - D) Charge density
42. The electric field is strongest:
- A) Where lines are close
 - B) Where lines are far
 - C) Uniform
 - D) None
43. For a negative point charge, field lines:
- A) Point outward
 - B) Point inward
 - C) Are tangential
 - D) Form loops
44. Electric field inside a conductor:
- A) Zero
 - B) Maximum
 - C) Uniform
 - D) Random

45. Electric field due to infinite line charge is proportional to:
- A) $1/r^2$
 - B) $1/r$
 - C) r
 - D) r^2
46. Which of these is a vector quantity?
- A) Electric potential
 - B) Electric field
 - C) Electric flux
 - D) Potential energy
47. Neutral point in dipole:
- A) $E = 0$
 - B) Maximum E
 - C) Minimum potential
 - D) High torque
48. Electric field at equatorial line of dipole is:
- A) Along dipole axis
 - B) Perpendicular to dipole axis
 - C) Zero
 - D) Radial
49. Electric flux is:
- A) Scalar
 - B) Vector
 - C) Tensor
 - D) None
50. The SI unit of electric flux is:
- A) $\text{N}\cdot\text{m}^2/\text{C}$
 - B) N/C
 - C) V/m
 - D) C/N